

EE 152 Project Report

Low-Light Image Enhancement Using Homomorphic Filtering

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Introduction:

Low-light images suffer from several degradations that make them difficult to interpret and process. These degradations include low brightness, reduced contrast, and noise amplification and uneven illumination caused by varying lighting conditions. Such issues negatively affect both human perception and the performance of automated computer vision systems, including surveillance systems, autonomous vehicles, and low-light photography pipelines. Therefore, developing robust enhancement techniques for low-light scenes is an important problem in image processing.

A commonly used model for understanding image formation is assuming that an image can be expressed as a product of two components: illumination and reflectance. Illumination represents low-frequency variations caused by lighting conditions, while reflectance represents the high-frequency structural details of the scene. Poor illumination dominates low-light images, making the underlying reflectance difficult to observe.

In this project, we implement and evaluate a homomorphic filtering-based low-light enhancement algorithm, which aims to suppress illumination while amplifying the reflectance. We compare this approach with traditional baseline enhancement techniques, including histogram equalization, plus median filtering and gamma correction. Because low-light datasets do not include bright ground-truth versions of images, we rely on no-reference image quality metrics like NIQE, BRISQUE, entropy, and Tenegrad. Experimental results demonstrate that parameter choices in the homomorphic filter influence image quality and how the method compares with baseline techniques.

Related Work:

Several classical and modern approaches have been proposed for low-light enhancement. Traditional contrast enhancement methods include Histogram Equalization, which redistributes pixel intensities to achieve a uniform histogram. While HE improves global contrast, it often amplifies noise and produces unnatural results, especially in very dark images. Variants such as Adaptive Histogram Equalization and CLAHE attempt to mitigate these issues, but may still produce artifacts in extremely low illumination.

Gamma Correction is another commonly used method, mapping input intensities using a nonlinear power-law transition. Although simple to implement, gamma correction cannot adapt to spatially varying illumination and may brighten noise or flatten details.

More advanced approaches include Retinex-based models, which estimate illumination and reflectance components. Multi-scale Retinex improves local dynamic range, but requires several Gaussian scales and careful parameter selection. These methods can also introduce color distortion.

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Recent developments in low-light enhancement involve deep learning techniques, particularly convolutional neural networks trained on paired or unpaired datasets. However, these models require significant computational resources and large training datasets, which fall outside the scope of this class.

Homomorphic filtering is a classic frequency-domain technique that explicitly models illumination and reflectance. By transforming the image into the log domain, illumination and reflectance become additive, enabling selective and frequency manipulation. Despite its age, this approach remains relevant due to its conceptual simplicity and effectiveness in certain applications of low-light situations. This project focuses on implementing homomorphic filtering, performing parameter analysis, and comparing it to baseline methods.

Technical Approach:

The Illumination-Reflectance Model:

We begin with the standard image formation model:

$$f(x,y) = i(x,y) * r(x,y)$$

i is illumination and r is reflectance

Taking the logarithm:

$$\ln f = \ln i + \ln r$$

Homomorphic Filtering Pipeline:

The filtering procedure consists of:

Convert the input image $f(x,y)$ to floating point and apply log:

$$g(x,y) = \ln(f(x,y) + \epsilon)$$

Compute the 2D fourier transform:

$$G(u,v) = F\{g(x,y)\}$$

Apply a modified high-pass filter:

$$H(u,v) = (\delta H - \delta L) \left(\frac{1}{1 + (D(u,v))^n} \right) + \delta L$$

where $\delta L < 1$ suppresses illumination, $\delta H > 1$ enhances reflectance, $D(u,v)$ is the distance from the frequency center, and n controls filter order.

Multiply frequency components:

$$S(u,v) = H(u,v) * G(u,v)$$

Inverse Fourier Transform and exponentiate:

$$f_{out}(x,y) = \exp(F^{-1} * \{S\})$$

Normalize to [0,1]. The final result brightens the image while revealing structural details.

Baseline Enhance Methods:

Histogram Equalization:

HE Improves global contrast but amplifies noise significantly in low-light environments. In this project, we additionally apply a median filter to these outputs:

$$I_{HE+Med} = \text{Median Filter}(I_{HE})$$

To suppress noise artifacts.

Gamma Correction applies $I_{out} = I_{in}^{\gamma}$. With $\gamma > 1$, brightening dark regions.

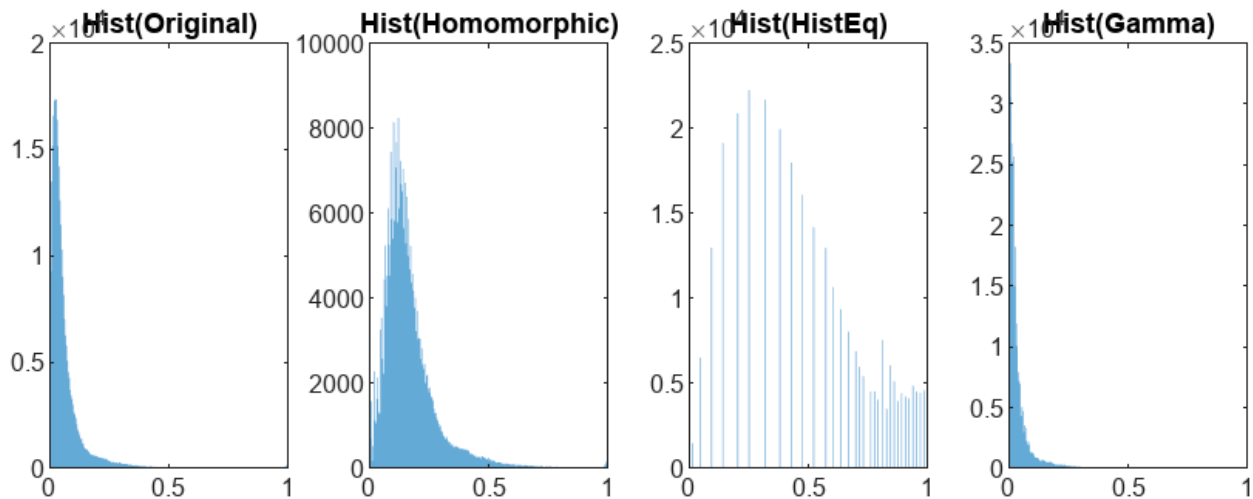
We use a subset of the ExDark dataset, a real-world low-light image collection. Several categories, including people and street scenes, were used for testing. Additional synthetic low-light images were produced from standard grayscale images to validate controlled performances. We have evaluation metrics that are no-reference quality. NIQE has lower values giving more natural images. BRISQUE is when lower indicates better quality. Entropy measures information content and Tenengrad sharpness measures edge strength.

Experimental Results:

Visual Comparison:

- HE dramatically boosts brightness but introduces strong noise
- HE + Median Filtering reduces noise but blurs edges
- Gamma Correction moderately brightens images but fails on uneven illumination
- Homomorphic filtering produces smoother illumination, preserves edges, and avoids excessive noise amplification

Parameter Study:



NIQE, BRISQUE, Entropy, and Tenengrad:

Homomorphic:

3.7000 25.9159 6.1747 0.0018

HistEq:

4.1543 22.1065 4.5759 0.0023

Gamma :

4.0077 30.3352 4.7771 0.0014

We experimented with:

- Gamma L from [0.2,0.6]
- Gamma H from [1.8,3.0, 0.5]
- Do from [30,100,250]
- Filter Order n from [1,3]

Results Show:

- Low Gamma L improves brightness but can wash out details
- High Gamma H enhances sharpness but may increase noise
- Higher Do improves enhancement consistency
- Lower order n produces smoother transitions and avoids ringing artifacts

Optimal parameters were found to be:

Gamma L = 0.6 Gamma H = .5 Do = 250 n = 1

Method	NIQE ↓	BRISQUE ↓	Entropy ↑	Tenengrad ↑
Homomorphic	improved vs original; competitive vs baselines	moderate	high	strong
HistEq	significantly worse (high noise)	poor	very high	unstable
HistEq + Median	improved BRISQUE but reduced sharpness	good	moderate	low
Gamma	reasonable performance	moderate	high	lower than homomorphic

(I got the table layout from ChatGPT)

Observations:

- Homomorphic filtering balances noise suppression and detail enhancement
- HistEq + Median improves perceptual quality but sacrifices sharpness
- Gamma Correction is consistent but limited
- Homomorphic Filtering yields the best trade-off overall

Conclusion:

This project implemented a low-light enhancement system using homomorphic filtering and compared it to traditional baseline methods. By modeling illumination and reflectance separately, the homomorphic filter effectively improves brightness and detail visibility while limiting noise amplification. Parameter tuning played a critical role in achieving optimal performance. Both qualitative and quantitative evaluations demonstrated that the homomorphic filter outperformed histogram equalization and gamma correction in achieving natural-looking enhancement. Future work includes extending the method to color images, implementing multi-scale filtering, and exploring learning based enhancement models.

References:

R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed., Pearson, 2018.

D. J. Jobson, Z. U. Rahman, and G. A. Woodell, "A Multiscale Retinex for Bridging the Gap Between Color Images and the Human Observation of Scenes," *IEEE Transactions on Image Processing*, vol. 6, no. 7, 1997.

A. Mittal et al., "Making a 'Completely Blind' Image Quality Analyzer," *IEEE Signal Processing Letters*, vol. 20, no. 3, 2013.

ExDark Dataset – <https://github.com/cs-chan/ExDark>